VILLAGE DEVELOPMENT PLATFORM

Relational Database Schema

A structured schema for the Community Services, Personal Services, and Agriculture modules.

Document purpose	Defines the main relational tables, primary keys, foreign keys, and important constraints for the MVP.
Database approach	Relational database design; each table represents an entity or a relationship that needs its own attributes.
Key conventions	PK = Primary Key; FK = Foreign Key; UNIQUE = values must not repeat; timestamps use a consistent database datetime type.

Module coverage

1. Community Services — citizen complaints, department prediction, evidence, and status history.
2. Personal Services — provider profiles and customer service requests.
3. Agriculture — products, local shops, and shop inventory.

Implementation note: this is the proposed MVP schema. Confirm it against your final requirements before creating database migrations.

Relational Tables and Attributes

The tables below describe the proposed fields and constraints. Field data types should be selected during implementation based on the chosen database engine.

1. USER

Field	Description	Key / constraint
user_id	Primary identifier	PK
full_name	User's full name	
phone	Phone number used for account contact/login	UNIQUE
password_hash	Secure password hash; never store a plain-text password	
role	MVP role, e.g. citizen, service_provider, farmer, shopkeeper, government_official, admin	
preferred_language	Preferred interface language	
created_at	Account creation timestamp	

2. DEPARTMENT

Field	Description	Key / constraint
department_id	Primary identifier	PK
department_name	Department name	UNIQUE
description	Short description of department responsibilities	
contact_details	Public contact information	

3. COMPLAINT

Field	Description	Key / constraint
complaint_id	Primary identifier	PK
citizen_id	User who submitted the complaint	FK → USER.user_id
description	Complaint text	
latitude	Latitude of reported issue; nullable if unavailable	
longitude	Longitude of reported issue; nullable if unavailable	
status	Current complaint status	
predicted_department_id	Department predicted by the ML classifier	FK → DEPARTMENT.department_id
ml_confidence	Classifier confidence score, if available	
created_at	Complaint creation timestamp	

3A. COMPLAINT — workflow design note

Field	Description	Key / constraint
Recommended extension	Consider adding assigned_department_id separately from predicted_department_id if officials can review or reassign a complaint.	DESIGN NOTE
Reason	The ML-predicted department and the department actually handling the complaint may differ.	DESIGN NOTE

4. COMPLAINT_EVIDENCE

Field	Description	Key / constraint
evidence_id	Primary identifier	PK
complaint_id	Complaint associated with the uploaded file	FK → COMPLAINT. complaint_id
file_url	Storage URL or path for the photo/document	
file_type	MIME type or file category	
uploaded_at	Upload timestamp	

5. COMPLAINT_STATUS_HISTORY

Field	Description	Key / constraint
history_id	Primary identifier	PK
complaint_id	Complaint whose status changed	FK → COMPLAINT. complaint_id
changed_by	User who made the status change	FK → USER.user_id
old_status	Status before the change; nullable for initial status	
new_status	Status after the change	
note	Optional explanation or update	
changed_at	Timestamp of the status change	

6. SERVICE_PROVIDER_PROFILE

Field	Description	Key / constraint
provider_user_id	Provider account identifier; also references USER	PK, FK → USER.user_id
service_category	Type of service offered	
service_description	Description of services offered	
service_area	Village, locality, or area served; structure can be refined later	
is_verified	Whether the provider has been verified	

7. SERVICE_REQUEST

Field	Description	Key / constraint
request_id	Primary identifier	PK

Field	Description	Key / constraint
customer_id	User who requested the service	FK → USER.user_id
provider_user_id	Provider receiving the request	FK → SERVICE_PROVIDER_PROFILE.provider_user_id
service_description	Requested work and details	
service_location	Location where service is needed	
status	e.g. pending, accepted, rejected, in_progress, completed, cancelled	
requested_at	Request creation timestamp	
updated_at	Last update timestamp	
response_note	Optional provider response or reason for rejection	OPTIONAL

8. SHOP

Field	Description	Key / constraint
shop_id	Primary identifier	PK
owner_user_id	User account that owns/manages the shop	FK → USER.user_id
shop_name	Shop display name	
address	Shop address or village/locality	
latitude	Latitude, if available	
longitude	Longitude, if available	
phone	Shop contact number	
is_active	Whether the shop is active in the platform	

9. PRODUCT

Field	Description	Key / constraint
product_id	Primary identifier	PK
product_name	Product name	
category	e.g. fertilizer, pesticide, seed, or other agricultural input	
brand	Brand/manufacturer, if applicable	
unit	Unit used for stock quantity, e.g. kg, litre, packet	

10. SHOP_INVENTORY

Field	Description	Key / constraint
inventory_id	Primary identifier	PK
shop_id	Shop holding the product	FK → SHOP.shop_id

Field	Description	Key / constraint
product_id	Product being stocked	FK → PRODUCT.product_id
quantity_available	Current quantity available	
price	Listed price for the specified unit	
availability_status	e.g. in_stock, low_stock, out_of_stock	
last_updated	Last inventory update timestamp	
UNIQUE(shop_id, product_id)	Prevent duplicate inventory rows for the same product at one shop	CONSTRAINT

Relationships and Constraints

Relationship	Cardinality	Meaning
USER → COMPLAINT	1 : N	One user can submit multiple complaints.
DEPARTMENT → COMPLAINT	1 : N	One department may be predicted for many complaints.
COMPLAINT → COMPLAINT_EVIDENCE	1 : N	A complaint can have multiple photos/documents.
COMPLAINT → COMPLAINT_STATUS_HISTORY	1 : N	A complaint can have multiple status changes.
USER → SERVICE_REQUEST (customer)	1 : N	One customer can make multiple service requests.
SERVICE_PROVIDER_PROFILE → SERVICE_REQUEST	1 : N	One provider can receive multiple requests.
USER → SHOP (owner)	1 : N	A user can manage one or more shops, if permitted.
SHOP → SHOP_INVENTORY	1 : N	A shop can list many products.
PRODUCT → SHOP_INVENTORY	1 : N	A product can be listed by many shops.

Important Design Rules

• **Primary keys:** every table has a stable primary key; SERVICE_PROVIDER_PROFILE uses provider_user_id as both PK and FK to USER.

• **Foreign keys:** use database constraints to prevent references to missing users, complaints, departments, providers, shops, or products.

• **Unique constraints:** USER.phone and DEPARTMENT.department_name are unique; SHOP_INVENTORY enforces one row per shop/product pair.

• **Passwords:** store only secure password hashes, never plain-text passwords.

• **ML output:** store the predicted department and, where supported, a confidence score. Do not treat a model prediction as a final official assignment automatically.

• **Derived values:** provider average ratings are excluded from the MVP schema because ratings/reviews are not yet in scope.

• **Role design:** a single USER.role is simple for the MVP. If one person must hold multiple roles, introduce a USER_ROLE mapping table later.

Suggested Next Steps

1. Review this schema against the ER diagram and the required project specification.

2. Choose the database engine and assign data types, defaults, nullability, and deletion/update rules.

3. Create migrations or SQL DDL and test primary-key, foreign-key, unique, and check constraints.

4. Build the Community Services module first, including complaint status history.

5. Add the ML classifier as a separate service/module that returns a predicted department and confidence score.